

Cole Schreiner

🔗 Checkout my biggest projects at: [UserIsMissing.GitHub.io](https://UserIsMissing.github.io)
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About Me

I am a Robotics Engineering graduate from UC Santa Cruz with 10+ years of hands-on experience in microelectronics, embedded systems, and automation. I bring a strong proficiency in microcontroller programming, software development, and CAD-based mechanical design, supported by a track record of leading small technical teams to deliver unique engineering solutions. I thrive in multidisciplinary environments and am driven to build innovative, novel technologies. I'm excited to apply my technical versatility to forward-thinking robotics organizations.

Education

Univ. of California, Santa Cruz **Grad. 2026**
Robotics Concentration *Grad. w/ Honors*

- M.S. Electrical & Computer Engineering
- B.S. Robotics Engineering
- Minor in Electrical Engineering

Work Experience

UC Santa Cruz **Spring 2026**
Mechatronics Teacher's Assistant

- Led 108 students to compete in a robotics competition
- Taught Electrical, Mechanical, and Software principles
- Managed 30 teams of students to complete engineering labs on schedule

Technical Skills

Languages: C, C++, Java, Python, Verilog, MATLAB

Tools and Programs: AutoDesk, OnShape, SolidWorks, Git, Vivado, LTSpice, Oscilloscopes, 3D Printing, Laser Cutting

Microcontrollers: ESP32, Raspberry Pi, PIC32, Arduino

Circuit Theory: Kirchhoff's Laws, voltage and current division, op-amps, RL and RLC circuits, frequency response, filters, Bode plots, Thevenin/Norton equivalence, analog electronics, and PSpice circuit modeling

Data Structures: Trees, Heaps, Binary Search Trees, Graphs, BFS, and DFS

Engineering: Statics, Dynamics and kinematic motion of robots, Feedback Control Systems, VLSI, Microcontroller System Design, PyTorch, Imitation Learning

Major Projects

UAV AI Autopilot and Simulation - [\[Project Link\]](#) **February, 2026**

- **Objective:** Reengineering traditional UAV autopilot systems by replacing successive loop closure (nested PID) with a Recurrent Neural Network (RNN) inference engine using behavioral cloning.
- **Technical Expertise:** Developed a PyTorch-based imitation learning model to map 15 input features—including attitude, body-frame velocities, and airflow—to continuous control outputs for Throttle, Aileron, Elevator, and Rudder.
- **Implementation:** Integrated the neural controller into the university's proprietary framework by overriding control command logic to achieve autonomous path-following and stability under simulated wind perturbations.
- **Optimization:** Applied input normalization and Mean Squared Error (MSE) minimization against expert teacher data to validate the efficacy of a single "brain" handling complex flight physics in real-time.
- **Tools Utilized:** Python, PyTorch, Flight Simulation GUI, University Control Framework

Capstone - Wet-Dry Cycler (RNA Replicator) - [\[Project Link\]](#) **2024 - 2025**

Project Director - Origins of Life Simulation Device

- **Objective:** Designed an automated device to simulate prebiotic wet-dry cycling, aiding RNA/DNA polymerization research and enabling cost-effective RNA synthesis for pharmaceutical applications (e.g., siRNA production)
- **Technical Expertise:** Microcontroller programming, feedback control, fluid dynamics, thermal systems, and mechanical design
- **Software Proficiency:** GUI development, automation control, and real-time data analysis
- Collaborated with biomolecular researchers to advance RNA transcription and polymerization, with potential applications in RNA-based therapies such as siRNA

Professor Piano - [\[Project Drive\]](#) **March, 2025**

- Designed and developed a wearable piano teaching glove featuring piezoelectric sensors, real-time peak detection, and personalized calibration
- Integrated inertial measurement units (accelerometers and gyroscopes) to capture and analyze user movement, utilizing Euler angles for precise orientation tracking
- Led the integration of I2S audio playback for high-quality sound feedback, enhancing the user experience with immersive, real-time audio feedback